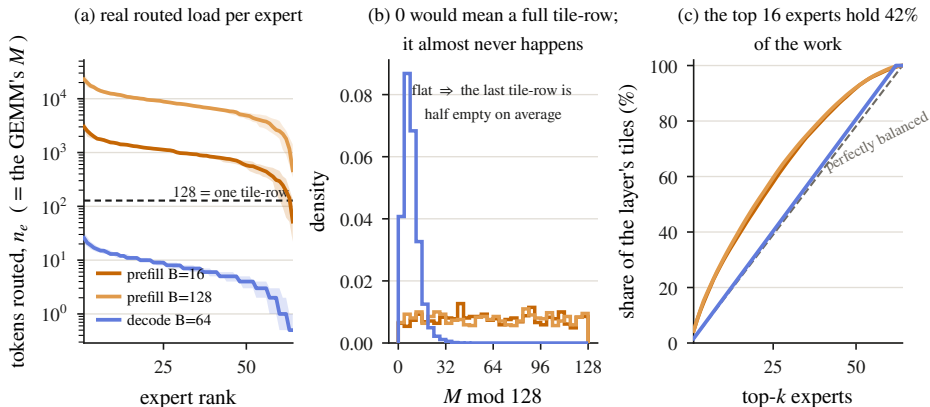




allenai/OLMoE-1B-7B-0924, 16 layers, 64 experts, top-8, on wikitext-103. Router hooked at the gate and top- k recomputed exactly as the block does it, so these are the model's own routing decisions, not a synthetic skew. Bands in (a) are the 10th-90th percentile across all layer-samples. An expert's GEMM is $(n_e \times 2048) @ (2048 \times 1024)$, so n_e is M ; tiles = $\lceil n_e / 128 \rceil \times 8$. Decode at B=64 gives 97% of experts a single tile-row -- their GEMMs are too small to fill even one wave.